

数值分析第十次作业

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1. 设 $B_n(x)$ 为 Bernoulli 多项式, 求证:

$$B_n(x) = (-1)^n B_n(1-x), \quad \forall x \in \mathbb{R}, \quad \forall n \geq 0.$$

解答.

对 $n \geq 0$ 归纳. 当 $n = 0, 1$ 时, 命题显然成立. 假设命题对 $n-1$ 成立 ($n \geq 2$), 对 n : 注意到

$$\begin{aligned} B_n(1-x) &= \int_0^{1-x} B_{n-1}(s) \, ds + B_n(0) \\ &= - \int_{1-x}^1 B_{n-1}(s) \, ds + B_n(0) \\ &= - \int_{1-x}^1 (-1)^{n-1} B_{n-1}(1-s) \, ds + B_n(0) \\ &= (-1)^n \int_0^x B_{n-1}(s) \, ds + B_n(0) \\ &= (-1)^n (B_n(x) - B_n(0)) + B_n(0) \\ &= (-1)^n B_n(x) + B_n(0)(1 - (-1)^n), \end{aligned}$$

这里用到了 $B_n(x), B_{n-1}(x)$ 的定义以及归纳假设, 故只需证: $B_n(0)(1 - (-1)^n) = 0$.

(1) 当 n 为偶数时, 显然成立.

(2) 当 n 为奇数时, 只需证 $B_n(0) = 0$. 注意到

$$0 = \int_0^1 B_n(x) \, dx = \int_0^1 \left(\int_0^x B_{n-1}(s) \, ds + B_n(0) \right) dx = B_n(0) + \int_0^1 B_{n-1}(s)(1-s) \, ds,$$

并且

$$\begin{aligned} \int_0^1 B_{n-1}(s)(1-s) \, ds &= \int_0^1 B_{n-1}(1-s)s \, ds \\ &= \int_0^1 (-1)^{n-1} B_{n-1}(s)s \, ds \\ &= \int_0^1 B_{n-1}(s)s \, ds \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 B_{n-1}(s) \, ds - \int_0^1 B_{n-1}(s)(1-s) \, ds \\
&= - \int_0^1 B_{n-1}(s)(1-s) \, ds,
\end{aligned}$$

即

$$\int_0^1 B_{n-1}(s)(1-s) \, ds = 0,$$

故 $B_n(0) = 0$.

从而命题对 n 也成立, 由数学归纳法知命题得证. □

2. 定义 $B_n(x)$ 从 $[0, 1]$ 到 $\mathbb{R}$ 的周期扩张 $\tilde{B}_n(x) : \mathbb{R} \rightarrow \mathbb{R}$ 为

$$\begin{cases} \tilde{B}_n(x) = B_n(x), & x \in [0, 1], \\ \tilde{B}_n(x+1) = \tilde{B}_n(x), & x \in \mathbb{R}. \end{cases}$$

求证: $\forall n \geq 2$, $\tilde{B}_n(x)$ 有 Fourier 级数展开式:

$$\begin{aligned}
\tilde{B}_{2m}(x) &= 2(-1)^{m-1} \sum_{k=1}^{+\infty} \frac{\cos 2\pi kx}{(2\pi k)^{2m}}, \\
\tilde{B}_{2m+1}(x) &= 2(-1)^{m-1} \sum_{k=1}^{+\infty} \frac{\sin 2\pi kx}{(2\pi k)^{2m+1}}.
\end{aligned}$$

解答.

对 n 归纳: 当 $n=2$ 时, 直接计算可得

$$\begin{aligned}
\int_0^1 B_2(x) \, dx &= \int_0^1 \left(\frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{12} \right) \, dx = 0, \\
\int_0^1 B_2(x) \cos 2\pi kx \, dx &= \int_0^1 \left(\frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{12} \right) \cos 2\pi kx \, dx = \frac{1}{(2\pi k)^2}, \\
\int_0^1 B_2(x) \sin 2\pi kx \, dx &= \int_0^1 \left(\frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{12} \right) \sin 2\pi kx \, dx = 0,
\end{aligned}$$

故

$$\tilde{B}_2(x) = 2 \sum_{k=1}^{+\infty} \frac{\cos 2\pi kx}{(2\pi k)^2},$$

命题成立. 假设命题对 $n-1$ 成立 ($n \geq 3$), 对 n : 注意到

$$\int_0^1 B_n(x) \, dx = 0,$$

$$\begin{aligned}\int_0^1 B_n(x) \cos 2\pi kx \, dx &= \frac{1}{2\pi k} \int_0^1 B_n(x) \, d(\sin 2\pi kx) = \frac{-1}{2\pi k} \int_0^1 B_{n-1}(x) \sin 2\pi kx \, dx, \\ \int_0^1 B_n(x) \sin 2\pi kx \, dx &= \frac{-1}{2\pi k} \int_0^1 B_n(x) \, d(\cos 2\pi kx) = \frac{1}{2\pi k} \int_0^1 B_{n-1}(x) \cos 2\pi kx \, dx,\end{aligned}$$

这里用到了分部积分和 $B_n(0) = B_n(1)$.

(1) 当 n 为偶数时, 设 $n = 2m$, 由归纳假设可得

$$\begin{aligned}\int_0^1 B_{2m-1}(x) \sin 2\pi kx \, dx &= \frac{(-1)^m}{(2\pi k)^{2m-1}}, \\ \int_0^1 B_{2m-1}(x) \cos 2\pi kx \, dx &= 0,\end{aligned}$$

故

$$\begin{aligned}\int_0^1 B_{2m}(x) \cos 2\pi kx \, dx &= \frac{(-1)^{m-1}}{(2\pi k)^{2m}}, \\ \int_0^1 B_{2m}(x) \sin 2\pi kx \, dx &= 0,\end{aligned}$$

从而

$$\tilde{B}_{2m}(x) = 2(-1)^{m-1} \sum_{k=1}^{+\infty} \frac{\cos 2\pi kx}{(2\pi k)^{2m}}.$$

(2) 当 n 为奇数时, 设 $n = 2m + 1$, 由归纳假设可得

$$\begin{aligned}\int_0^1 B_{2m}(x) \sin 2\pi kx \, dx &= 0, \\ \int_0^1 B_{2m}(x) \cos 2\pi kx \, dx &= \frac{(-1)^{m-1}}{(2\pi k)^{2m}},\end{aligned}$$

故

$$\begin{aligned}\int_0^1 B_{2m+1}(x) \cos 2\pi kx \, dx &= 0, \\ \int_0^1 B_{2m+1}(x) \sin 2\pi kx \, dx &= \frac{(-1)^{m-1}}{(2\pi k)^{2m+1}},\end{aligned}$$

从而

$$\tilde{B}_{2m+1}(x) = 2(-1)^{m-1} \sum_{k=1}^{+\infty} \frac{\sin 2\pi kx}{(2\pi k)^{2m+1}}.$$

因此命题对 n 也成立, 由数学归纳法知命题得证. □